# Daniel Gerardo Agüero Agüero

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*Full-Stack Software Engineer with hands-on experience building scalable applications using C#, ASP.NET Core, TypeScript, and React. Proven track record in developing data-driven interfaces, RESTful APIs, and SQL database models. Strong focus on SOLID principles, maintainable software architecture, automated testing, and high-quality code delivery.*

## TECHNICAL SKILLS

**Programming Languages:** JavaScript, TypeScript, Python, C#, Java
**Frontend:** React, Angular, Tailwind, Nextjs, TanStack Query, Zustand
**Backend:** Node.js, Express, .NET, FastAPI, Spring
**Databases:** MySQL, PostgreSQL (Neon), SQL, Prisma ORM
**Cloud & Deployment:** Vercel, Railway, Supabase, Appwrite
**Testing & Quality:** Vitest, Unit Testing, Integration Testing, Zod, Sentry
**Tools & Practices:** Git, GitHub, OpenCode (ChatGPT, Claude Code, Kimik3), CI/CD, Power BI

## PROFESSIONAL EXPERIENCE

**Software Engineering Intern / SITSA / Quizy** ***April 2026 – August 2026***
Full-Stack Developer & QA

- **Developed** a production-ready examination management system using **C# and .NET**, implementing authentication, user management, academic subjects, student portals, examinations, and grading workflows.
- **Designed** and implemented complex examination workflows with **server-side state persistence**, including randomized question ordering, answer persistence, exam recovery after connection/browser failures, and server-authoritative time tracking.
- **Implemented** authentication and credential recovery for administrators, teachers, and students, including session management, password recovery, and secure password updates.
- **Applied dependency injection and SOLID principles** to decouple services and enable parallel development through mocked service implementations.
- **Developed** database operations using **SQL Stored Procedures with parameterized inputs**, implementing standardized server responses and structured error handling.
- **Ensured** software quality through **135+ unit and integration tests** covering happy paths and edge cases across core modules.
- **Collaborated** in a two-developer team using Git-based workflows, weekly merges, agile feedback cycles, and technical documentation for database procedures and application functions.

**Financial SaaS / TAKU-Cash / takucash.com** ***June 2026 – Present***
Full-Stack Software Engineer

- **Designed and developed** a financial SaaS independently using **TypeScript, React, Express, Prisma, and TiDB**, currently serving **26 active users**, with modules for accounts, transactions, budgets, categories, transfers, financial reporting, and subscription management.
- **Architected** the backend using **Hexagonal Architecture**, dependency injection, and separated domain/application/infrastructure/interface layers, enabling maintainable and decoupled business logic.
- **Implemented** secure authentication and subscription workflows using **Clerk, OAuth 2.0, Paddle, and webhooks**, including webhook signature verification, rate limiting, account-enumeration protection, authorization, and automated premium subscription management.
- **Engineered** complex financial business rules for **multi-currency accounts and transactions**, including CRC/USD transfers, user-defined exchange rates, balance validation, transaction integrity, and currency normalization across financial reports.

- **Ensured** application reliability through approximately **220 unit and integration tests**, Sentry monitoring, standardized API responses/error handling, and comprehensive technical documentation; deployed the frontend on Vercel and backend on Railway.

## FREELANCE SOFTWARE DEVELOPMENT

**Livestock & Farm Management Platform / Livestockpro /** ***May 2026 – June 2026***
**livestockpro.com**
Full-Stack Developer

- **Developed a web-based livestock management platform** that centralized the administration of farms, livestock, breeds, and financial transactions through a full-stack solution built with **React, TypeScript, Express, Prisma, and MySQL**.
- **Designed and implemented a scalable architecture** based on **dependency injection** and layered separation (**Routes → Middlewares → Controllers → Services → Database**), improving maintainability, extensibility, and system decoupling.
- **Implemented role-based access control (RBAC)**, **JWT authentication**, and route protection across both frontend and backend applications, ensuring secure access to critical business operations.
- **Built interactive dashboards and analytical reports** featuring advanced filters, financial metrics, and data visualizations to monitor livestock inventory, expenses, revenue, and net profit in real time.
- **Ensured system quality and reliability** through **200+ automated unit and integration tests** with **Vitest**, type-safe validation using **Zod**, and production deployments on **Railway and Vercel**.

**Swine Mortality Management System / PostDestete / postdestete.com** ***December 2025 – January 2026***
Full-Stack Developer

- **Developed a swine mortality analytics platform** for a farm with 2,000+ animals, **implementing a filter-first pattern** on the frontend and **optimized aggregated queries** on the backend with Prisma (including dynamic time-granularity groupings and zero-filled categories).
- **Built a SPA** using React 19, TypeScript, Tailwind CSS, and TanStack Query v5, **optimizing selective cache invalidation** and **using Zustand for derived state projections** without redundant network requests.
- **Designed a secure authentication system** based on short-lived JWTs in httpOnly cookies, **hashed refresh token rotation**, **CSRF protection** (checkOrigin middleware), **IP-based rate limiting**, and **bcrypt password encryption** (12 rounds).
- **Implemented a layered architecture** with Dependency Injection (DI) in Express 5 and strict Zod validation on both ends, **backed by a suite of 199 integration tests** using Vitest and Supertest against an isolated database.

## EDUCATION

**Technical University of Costa Rica (Universidad Técnica Nacional)** **Alajuela, Costa Rica**
Bachelor's Degree in Software Engineering *January 2023 – April 2026*

## LANGUAGES

- Spanish — Native
- English — C1 Advanced (EF SET Certified) Certified